

# Kusto (KQL) Prioritized Features

Work in progress...

This page lists Kusto features as a whole and is not limited specifically to the beta. We would select features from this list to include in the beta that would provide the highest impact.

Impact = how much will it impact users (how delightful will it make them) and how many users will it reach

Effort = how difficult it is to implement

The impact value is chosen based on data about the typical historical searches run by customers, gathered either from out-of-the-box content or from the field (professional services).

There are a few features in this table that are must-haves, i.e. essential features that if not present, the customer won't even consider our product as a solution to their problem. I'm thinking those could be broken out into their own table to distinguish them.

Please note that most of the examples here do not strictly follow our current base/default schema so you might see references to strange columns, it's simply meant to convey the use case.

This list isn't complete, it's being populated. Kusto has so many features, I'm still trying to get a hang of how "deep" this table should go, some of these items are very very basic but some are complex, and I'm sure some of the basic things are already implemented from the POC.

| Feature # | Be able to:                           | Impact (1-6, 6=most impact) | Effort (1-6, 6=most difficult)   | In AQL? | Implementation Spec                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Examples                                                                                                                                                                                                                                                   | Dev Notes                                                                                                                                                                                                                              |
|-----------|---------------------------------------|-----------------------------|----------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | Limit returned results                | 6                           | 2                                | Yes     | <ul style="list-style-type: none"><li>Return up to the specified number of rows, using the <code>take</code> table operator.</li><li>Must implement the <code>limit</code> table operator as an alias to <code>take</code>.</li><li>Does not deviate from KQL <a href="#">implementation</a>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                            | maps to limit operator on CH                                                                                                                                                                                                           |
| 2         | Select column(s) to display           | 6                           | 2 for basic<br>5 for expressions | Yes     | <ul style="list-style-type: none"><li>Select the columns to display in the output, in the specified order, using the <code>project</code> operator.</li><li>Does not have to implement renaming columns. That's covered in feature #13.</li><li>Does not have to implement supporting expressions to insert new columns, that's covered in another feature.</li><li>Deviates from KQL <a href="#">implementation</a> for now until the above referenced variants are implemented.</li></ul>                                                                                                                                                                                                                                                      | Show me the starttime and eventname columns:<br><br><code>events</code><br><code>  project starttime, eventname</code>                                                                                                                                     | Basic function Maps to column listing on CH.<br><br>Expression support should be called out as a supported alternative example<br><code>  project a=col1</code><br><br><code>  project a=1+5</code><br><code>  a= tolower(col1)</code> |
| 3         | Sort the output rows                  | 6                           | 2                                | Yes     | <ul style="list-style-type: none"><li>Sort the output rows into order by one or more columns, using the <code>sort by</code> table operator.</li><li>Must implement the <code>order by</code> table operator as an alias to <code>sort by</code>.</li><li>Must allow specifying ascending and descending order using <code>asc</code> or <code>desc</code>.</li><li>If the direction is not specified, default to <code>desc</code>.</li><li>Must allow specifying the direction for each column separately.</li><li>Must allow specifying where to place nulls values; at the beginning using <code>nulls first</code> or at the end using <code>nulls last</code>.</li><li>Does not deviate from KQL <a href="#">implementation</a>.</li></ul> | Sort the results by the username descending:<br><br><code>events</code><br><code>  sort by username desc</code><br><br>Sort the results by the username and then computername:<br><br><code>events</code><br><code>  sort by username, computername</code> |                                                                                                                                                                                                                                        |
| 4         | Perform numerical operations          | 4                           | 4                                | Yes     | <ul style="list-style-type: none"><li>Must allow add, subtract, multiply, divide, modulo, less than (or equals), more than (or equals), equality (equals and not equals) on numerical data types.</li><li>It seems as though different <a href="#">programming languages treat modulo operations with negative numbers differently</a>. For now, we can implement whatever CH does with negatives.</li><li>Not sure how to deal with floating numbers. For example, does divide always return a float type? I believe QRadAr works that way, so we could do it like that.</li><li>Depending on the above case about the modulo operation, this could deviate from the KQL <a href="#">implementation</a>.</li></ul>                              |                                                                                                                                                                                                                                                            | Because these can be used in many different places within a query, corner cases may be difficult to cover initially.                                                                                                                   |
| 5         | Perform string equality operations    | 6                           | 2                                | Yes     | <ul style="list-style-type: none"><li>Must allow equals, equals case-insensitive, not equals, and not equals case-insensitive operations on string data types, using <code>==</code>, <code>=~</code>, <code>!=</code>, and <code>!=~</code> operators.</li><li>Does not deviate from KQL <a href="#">implementation</a>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                            | Map to string equality and like/like functions in CH                                                                                                                                                                                   |
| 6         | Filter using common string operations | 5                           | 2                                | Yes     | <ul style="list-style-type: none"><li>Must allow filtering data on string data types using the <code>contains</code>, <code>startswith</code>, and <code>endswith</code> operators.</li><li>Must support their case insensitive <code>contains_cs</code>, <code>startswith_cs</code>, and <code>endswith_cs</code> operators.</li><li>Must support their negating <code>!contains</code>, <code>!startswith</code>, and <code>!endswith</code> operators.</li><li>Must support their negating case insensitive <code>!contains_cs</code>, <code>!startswith_cs</code>, and <code>!endswith_cs</code> operators.</li><li>Does not deviate from KQL <a href="#">implementation</a> (listed in the table <a href="#">here</a>).</li></ul>           |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                        |

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| 7  | Filter using a list of elements           | 6 | 4                                                                                                                           | Yes   | <ul style="list-style-type: none"> <li>Filter data on string or numerical data types using a comma-separated list of elements, using the <code>in</code> operator.</li> <li>Must implement the case insensitive <code>in~</code>, negating <code>!in</code>, and negating case insensitive <code>!in~</code> operators.</li> <li>This feature item does not include supporting subselect.</li> <li>Deviates from the KQL <a href="#">implementation</a> until we implement the tabular expression (subselect) feature.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Show me all powershell processes:<br><br>events<br>  where process in~ ("powershell.exe", "powershell_ise.exe")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Overlap with 'in (subselect)' Requires a lot of testing                                                                                                              |
| 8  | Filter using a regex value                | 6 | 4                                                                                                                           | Yes   | <ul style="list-style-type: none"> <li>Filter data on string data types using a regular expression.</li> <li>There are different regex engines (POSIX basic, POSIX extended, PCRE, etc.); ideally, what we use needs to be feature rich, so not just POSIX basic or extended, but more akin to PCRE for example.</li> <li>It looks like the CH <a href="#">match function</a> uses <a href="#">re2</a>. I haven't heard of it before but it looks pretty extensive, so that should suffice.</li> <li>Does not deviate from the KQL <a href="#">implementation</a>. In our final public documentation, we should document which regex engine we use so users know it.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Catch an impacket credential dump attack:<br><br>events<br>  where EventID == "5145"<br>  where ShareName matches regex "\\.\.*\\ADMIN\\\$"<br>  where RelativeTargetName matches regex "SYSTEM32\\.\\.tmp"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mostly testing of regex syntax to see if there are mis-matches or lack of support in CH. Should be able to use the 'match(haystack, pattern)' function in clickhouse |
| 9  | Filter numerical data types using a range | 2 | 3                                                                                                                           | Yes   | <ul style="list-style-type: none"> <li>Filter numerical data types matching the values in a range, using the between operator.</li> <li>Must implement the negating <code>!between</code> operator.</li> <li>Range is inclusive.</li> <li>Syntactic sugar for <code>col &gt;= # and col &lt;= # and NOT col &gt;= # and col &lt;= #</code>.</li> <li>Deviates from the KQL <a href="#">implementation</a> in that this is only for the numerical data type, not for datetime, that's implemented in feature #10. Once both of these are implemented, this would not deviate from the KQL implementation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                      |
| 10 | Filter date/time data types using a range | 2 | 5                                                                                                                           | Yes   | <ul style="list-style-type: none"> <li>Filter datetime data types matching the values in a range, using the between operator.</li> <li>Must implement the negating <code>!between</code> operator.</li> <li>Range is inclusive.</li> <li>Must allow user to enter a date in the format yyyy-MM-dd HH:mm:ss.</li> <li>Ideally, should allow user to use other date formats as well, but not a hard requirement.</li> <li>Must allow the right hand side to take an interval value such that the right hand side evaluates to left hand side + specified interval.</li> <li>Deviates from the KQL <a href="#">implementation</a> in that this is only for the datetime data type, not for numerical, that's implemented in feature #9. Once both of these are implemented, this would not deviate from the KQL implementation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | There are many functions and patterns to construct dates as well as short hand time-frames (example 8d)                                                              |
| 11 | Search faster using whole terms           | 3 | 2 for basic string ops (startswith endswith etc)<br><br>3 Basic list of terms<br><br>5 to include <i>tabular expression</i> | Maybe | <ul style="list-style-type: none"> <li>KQL has a set of "has" operators that operate on "terms" which is another word for "tokens" - a string value is broken down into sequences separated by non-alphanumeric characters (boundaries), that's stored in a separate index, read <a href="#">here</a> for more details.</li> <li>has is not the same as contains. has only works on this tokenized index. See the examples for more details.</li> <li>Must support the has operator, and the case insensitive <code>has_cs</code>, negating <code>!has</code>, and negating case insensitive <code>!has_cs</code> operators.</li> <li>The benefit of this feature would be to theoretically allow faster searching than using regex or contains, at the detriment of using more storage for the additional index. The storage impact would need to be evaluated. CH is already fast though, so I'm not sure how "faster" this feature would be and if it warrants the effort to implement it. This will likely be tabled for a bit.</li> <li>In the CH website, I see references to a <a href="#">tokenbf_v1 index</a> and references to two functions <a href="#">hasToken</a> and <a href="#">hasTokenCaseInsensitive</a> - I believe these are what map to these KQL has operators.</li> <li>I could see this operator causing confusion because of the potential of not finding data if the filter is not properly crafted or when the user doesn't know what they're looking for. Documentation should be able to alleviate that.</li> <li>Does not deviate from the KQL <a href="#">implementation</a>.</li> </ul> | <p>Given a string value of North America it would be tokenized into two tokens: North and America and stored in an index. Doing has "North" returns true. Doing has "Nor" returns false because Nor is not a token in North America. Doing contains "Nor" returns true.</p> <p>A real world example: show me all powershell events that could involve a download which might indicate an attack.</p> <p>Powershell logs contain all the commands users are running. Given a column called command, it could have any of these values in the logs:</p> <pre>(New-Object System.Net.WebClient). DownloadFile('https://badwebsite.com/badfile.exe') (New-Object System.Net.WebClient). DownloadString('https://badwebsite.com/badfile.exe')</pre> <p>This is a trimmed version of possible values; values in a column like this that stores the commands being run from Powershell could be incredibly long (e.g. 400 characters), and Powershell logs account for a very large volume of logs.</p> <p>The user wants to find all commands that contain WebClient. Typically, they would do this: where command <b>contains</b> "WebClient" - however, that might be slow, given how large the entire value of the column could be. But if the column was tokenized, doing a where command <b>has</b> "WebClient" might be faster. So this feature is purely related to search speed.</p> | Is the tabular expression to be included as well?<br><br>where col has_any ( <i>tabular expression</i> )                                                             |
| 12 | Filter using contains any or contains all | 3 | 2                                                                                                                           | No    | <ul style="list-style-type: none"> <li>has is not the same as contains. See feature #11 for more details.</li> <li>Expand the contains operator to allow using an "any" or "all" operation by supplying a list of elements, by implementing contains_any and contains_all operators.</li> <li>KQL does not have this feature.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Show me all powershell events that could involve a download.</p> <p>Without a contains_any function, the user would need to do this:</p> <pre>events   where command contains "webclient" or command contains "downloadfile" or command contains "downloaddata"</pre> <p>But with a contains_any function, it would be:</p> <pre>events   where command contains_any ("webclient", "downloadfile", "downloaddata")</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                      |

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| 13 | Rename columns using the project operator        | 2 | 2 | Yes | <ul style="list-style-type: none"> <li>Must allow the project operator to rename a column when selecting it to be displayed.</li> <li>This does not include evaluating expressions.</li> <li>Deviates from KQL <a href="#">implementation</a> for now. See feature #2 for more details.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Display the starttime column and the eventname column, but rename the starttime column to timestamp:</p> <pre>events   project timestamp=starttime, eventname</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Will map to select col as name ...                                                                                                                                                                                                                                                                  |
| 14 | Rename columns using the project-rename operator | 1 | 2 | Yes | <ul style="list-style-type: none"> <li>Rename columns using the project-rename operator.</li> <li>Does not deviate from the KQL <a href="#">implementation</a>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Display the starttime column and the eventname column, but rename the starttime column to timestamp:</p> <pre>events   project starttime, eventname   project-rename timestamp=starttime</pre> <p>Renaming can be done before the project as well:</p> <pre>events   project-rename timestamp=starttime   project timestamp, eventname</pre> <p>Renaming using both project-rename and project is legal and would follow the flow of the query.</p> <p>For example, this is allowed (the result is two columns: ts and eventname):</p> <pre>events   project timestamp=starttime, eventname   project-rename ts=timestamp</pre> <p>This is also allowed (the result is two columns: ts and eventname):</p> <pre>events   project-rename timestamp=starttime   project ts=timestamp, eventname</pre> <p>These two operators do not have to immediately follow each other, other operators could fall in between (given the piped nature of the language).</p> |                                                                                                                                                                                                                                                                                                     |
| 15 | Use base aggregate functions                     | 5 | 2 | Yes | <ul style="list-style-type: none"> <li>Must allow the following aggregation functions: <ul style="list-style-type: none"> <li>avg</li> <li>count</li> <li>dcount - it seems as though KQL's implementation allows passing an accuracy value, the same way CH's speaks about the count_distinct implementation - the default seems to use uniqExact which is an accurate function but requires more memory. If there's a way to specify the count_distinct implementation at query time, we could expose a similar feature. Otherwise, our implementation of dcount does not have to take the optional accuracy argument.</li> <li>min</li> <li>max</li> <li>stdev</li> <li>stdevp</li> <li>sum</li> <li>variance</li> <li>variancep</li> </ul> </li> <li>dcount deviates from the KQL implementation, all others do not.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>CH Aggregate functions <a href="https://clickhouse.com/docs/en/sql-reference/aggregate-functions/reference/count/">https://clickhouse.com/docs/en/sql-reference/aggregate-functions/reference/count/</a></p> <p>Only functions with equivalent in CH will be supported</p>                       |
|    | Use base aggregate functions with predicates     | 3 | 5 | Yes | avgif, countif, dcountif, minif, maxif, stdevif, sumif, varianceif                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | The function and predicate will need to be separated as CH does not have a combined func                                                                                                                                                                                                            |
|    | Aggregate by columns                             | 6 | 5 | Yes | Must allow one or more columns. Must allow specifying multiple aggregation functions (count, dcount, avg, etc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Show me all the source IPs for authentication failures for a user:</p> <pre>events   where username == 'john' and eventname='logonfailed'   summarize count() by sourceip</pre> <p>Show me authentication failures for a user by the combination of source IP and destination IP:</p> <pre>events   where username == 'john' and eventname == 'logonfailed'   summarize count() by sourceip, destinationip</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Full summarize support allows for</p> <pre>T   summarize [[Column =] Aggregation [, ...]] [by [Column =] GroupExpression [, ...]]</pre>                                                                                                                                                          |
|    | Work with time intervals                         | 5 |   | No  | Support understanding the timespan data type. Recognize time intervals, such as 2d, 1.5h, 10s.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Show me events that have a starttime from more than 3 days ago:</p> <pre>events   where starttime &gt;= now() - 3d</pre> <p>Alternatively, if ago() is supported, this would be replaced by starttime &gt;= ago(3d)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Time intervals is too vague. Can we enumerate all of the ones we need to support? For example <a href="https://docs.microsoft.com/en-us/azure/data-explorer/kusto/query/scalar-data-types/timespan">https://docs.microsoft.com/en-us/azure/data-explorer/kusto/query/scalar-data-types/timespan</a> |

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|  | Aggregate by time intervals            | 6 |   | Yes | Allow aggregating by time intervals using the bin function to allow intuitive time-based analytics.                                                                                                                                                                                                                | <p>Show me the number of logon failures every hour:</p> <pre>events   where eventname == 'logonfailed'   summarize count() by bin(starttime, 1h)</pre> <p>Show me the unique number of users exhibiting login success/failures every 2 hours. This could help me see trends and catch anomalies:</p> <pre>events   where eventname in ('logonsuccessful', 'logonfailed')   summarize dcount(username) by bin(starttime, 2h), eventname</pre>                                                                                                                                                                                                                                                            | Time intervals is too vague. Can we enumerate all of the ones we need to support? For this feature is it only 'by bin(<timecol>, <interval>)                                    |
|  | Filter using subquery                  | 5 |   | Yes | Related to matching a list (the in and !in operators)                                                                                                                                                                                                                                                              | <p>Show me all the source ips associated with the top 10 users that exhibited the most failed login attempts:</p> <pre>events   where username in (   events     where eventname == 'logonfailed'     summarize count() by username     sort by count desc     take 10)   summarize count() by sourceip</pre>                                                                                                                                                                                                                                                                                                                                                                                           | It should be stated, how deep the nesting of subqueries we want to be able to support. CH will impose default limits (100 levels) so we should state whether that is sufficient |
|  | Derive additional columns              | 4 |   | Yes | Create additional columns from a calculation or computation within the query.                                                                                                                                                                                                                                      | <p>Show me information about my VPN sessions:</p> <pre>events   where eventname == "VPN Session"   extend SessionDuration = EndTime - StartTime   extend SessionAge = now() - StartTime   project StartTime, EndTime, SessionDuration, SessionAge</pre> <p>Convert the throughput from megabits to gigabits:</p> <pre>events   extend throughput_gb = throughput_mb / 1024</pre> <p>Create a new column with just the hostname from a FQDN:</p> <pre>events   extend host = split(ComputerFQDN, ".")   [0]   project username, host</pre>                                                                                                                                                               |                                                                                                                                                                                 |
|  | Join (inner)                           | 4 |   | No  | Allow an inner join. Should also allow the option where I can specify different columns for left and right to join on.                                                                                                                                                                                             | <p>Catch Pass the Hash - MITRE T1550 - i.e. detect defense evasion and/or lateral movement - involves stringing two events together:</p> <pre>events   where TimeGenerated &gt;= ago(3d)   where EventID == 4624 and LogonType == 9   join kind=inner (   events     where TimeGenerated &gt;= ago(3d)     where EventID == 4648 and AccountType == "User" ) on LogonGuid   sort by TimeGenerated desc</pre>                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                 |
|  | Join (other flavors)                   | 1 |   | No  | Most of our use cases use inner join, so if it's difficult to implement other flavors of JOIN we can put this on the back burner.                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |
|  | Do free text search                    | 5 | 5 | Yes | Search the payload column.                                                                                                                                                                                                                                                                                         | search "john"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                 |
|  | Bind names to expressions (let)        | 6 | 6 | No  | Improves modularity and reuse, since this will let you break a potentially complex expression into multiple parts. This feature could help us implement views, where users can define them or we can ship them by default. For example, we could have a view for each kind of log source type, the view would have | <p>Catch Lateral Movement - MITRE TA0008 - Show me outlier users that are logging in to more computers than they normally do per day (i.e. hunting for abnormal behavior, using standard deviation):</p> <pre>let LogonsPerUserPerDay = events   where EventID == 4624   summarize ComputerCount=dcount(Computer) by Account, Day=bin(TimeGenerated, 1d);  let peakThreshold = LogonsPerUserPerDay   summarize AVG=avg(ComputerCount), STDEV=stdev(ComputerCount) by Account   project Account, STDEV, Threshold = AVG + STDEV, AVG;  LogonsPerUserPerDay   join kind=inner (peakThreshold) on Account   where ComputerCount &gt; Threshold   project Account, Day, ComputerCount, AVG, Threshold</pre> | This may need to be managed outside of clickhouse where queries could be substituted for variables before executing                                                             |
|  | Return an array of values in aggregate | 2 |   | No  | make_list and make_set. Similar to ARRAY_AGG in postgresql. I'll provide examples on why this is useful.                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |

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|  | Support materialized views |   |   |     | <a href="https://docs.microsoft.com/en-us/azure/data-explorer/kusto/management/materialized-views/materialized-view-overview">https://docs.microsoft.com/en-us/azure/data-explorer/kusto/management/materialized-views/materialized-view-overview</a> - essentially similar to QRadar's accumulator. This'll be on the back burner but will likely be revisited if we cannot meet search performance requirements. |                                                                                                                                         |
|  | Use base string functions  | 4 | 3 | Yes | indexof, isempty, isnull, split, strcat, strcat_delim, strlen, substring, toupper, tolower                                                                                                                                                                                                                                                                                                                         | indexof maps to ch position function. does not support an end index only start index<br><br>str_delim equivalent not directly supported |
|  | Encode/decode strings      | 3 | 2 | No  | base64 encode/decode, base16 (hex) encode/decode, url encode/decode                                                                                                                                                                                                                                                                                                                                                | base16 encode/decode not supported<br>URL encode not supported but encode is                                                            |